

ABSTRACT

Gesture-controlled robots are an emerging field in human-machine interaction where physical gestures are translated into control commands. This project aims to design and develop a two-wheel mobile robot that can be controlled through smartphone gestures using an accelerometer-based mobile application and an Arduino microcontroller. The system uses an HC-05 Bluetooth module for communication between the mobile device and the robot. The robot moves in different directions forward, backward, left, and right based on the tilt direction of the smartphone. This project demonstrates the integration of embedded systems, wireless communication, and robotics. The developed robot is low-cost, easy to build, and suitable for beginners in embedded systems and robotics.

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1. Introduction :

1.1 Overview :

Robotics has become an essential part of modern engineering, offering automation, safety, and efficiency across various sectors. Traditional robots require physical controls such as joysticks, remotes, or wired interfaces. However, with technological advancements in sensors and wireless communication, gesture-based control systems have gained popularity.

This project presents the design and implementation of a **gesture-controlled robot** using **Arduino UNO/Nano**, **HC-05 Bluetooth module**, and a **smartphone's inbuilt accelerometer**. The user controls the robot simply by tilting the smartphone, making the interface intuitive, user-friendly, and modern.

1.2 Motivation :

The major motivation behind this project is to explore natural human-machine interaction. Gesture control offers a hands-free, simple method for operating machines. This technology can be applied in:

- Robotics for disabled or elderly persons
- Hazardous environment navigation
- Smart automation
- Educational robotics

It reduces dependency on physical remotes and enhances user experience.

1.3 Problem Statement :

Existing robot control systems rely on traditional buttons or wired remotes which limit mobility and flexibility. There is a need for a wireless, intuitive, and easy-to-use control method that can operate robots remotely.

This project addresses the need for a low-cost, gesture-based control system using commonly available components.

1.4 Objectives :

The objectives of this project are:

1. To design a gesture-controlled robot using Arduino and Bluetooth communication.
2. To use smartphone accelerometer data for controlling robot movement.
3. To implement motor driving using the L293D driver module.

To create a simple, low-cost, and efficient wireless robotic system.

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2. System Design :

2.1 Block Diagram :

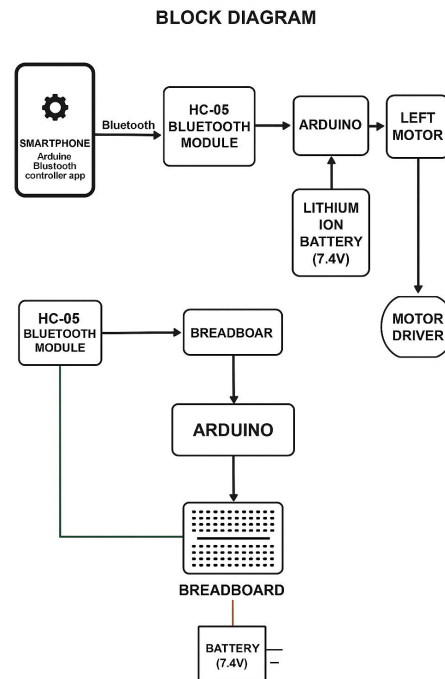


Figure 2.1: Block Diagram of Gesture-Controlled Robot Using Arduino

Transmitter Section (Smartphone):

- Uses a smartphone app to send control signals.
- Commands are transmitted via Bluetooth (Forward, Backward, Left, Right, Stop).
- No wired connection is required.

Receiver Section (Robot):

- HC-05 Bluetooth module receives the incoming commands from the smartphone.
- The received signals are passed to the Arduino, which processes and interprets the commands.
- Arduino sends control signals to the motor driver (L293D/L298N).
- The motor driver controls the left and right DC motors, enabling movement in the commanded direction.
- A 7.4V lithium-ion battery powers the Arduino, Bluetooth module, and motors.
- A breadboard is used for power distribution and connection support.

Overall Signal Flow:

Smartphone → HC-05 Bluetooth Module → Arduino UNO/Nano → Motor Driver (L293D/L298N) → DC Motors → Robot Chassis

2.2 Circuit Diagram :

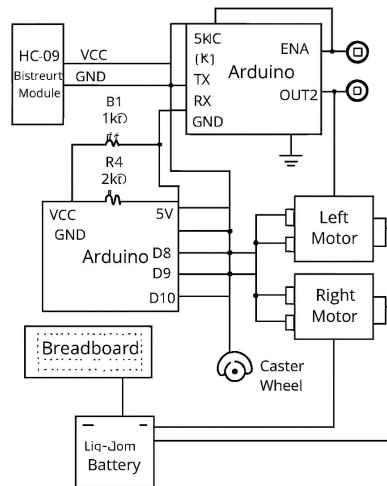


Figure 2.2: Circuit Diagram of Gesture-Controlled Robot Using Arduino

The circuit connects the Bluetooth module, Arduino, motor driver, and motors together for robot control.

HC-05 Bluetooth Module

- VCC → 5V
- GND → GND
- TX → Arduino RX (Pin 0)
- RX → Arduino TX (Pin 1 via voltage divider)

Motor Driver (L293D/L298N)

- IN1, IN2 → Arduino pins (Left motor)
- IN3, IN4 → Arduino pins (Right motor)
- Outputs → DC motors

Arduino

- Receives Bluetooth commands
- Sends control signals to motor driver

Power Supply

- Arduino powered via USB or 7.4V battery
- Motor driver powered using external battery
- Breadboard used for VCC/GND distribution

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This circuit ensures correct communication, safe voltage levels, and smooth robot movement.

2.3 Working Principle :

The gesture-controlled robot operates by using the smartphone's built-in accelerometer to detect the direction in which the phone is tilted. These tilt values are processed by the mobile application, which converts them into specific command characters. The entire working principle is explained step-by-step below:

1. Gesture Detection (Smartphone Accelerometer)

- The smartphone continuously monitors its orientation using the accelerometer sensor.
- When the user tilts the phone **forward, backward, left, or right**, the accelerometer values change accordingly.
- The mobile app interprets these changes and assigns a movement command for each gesture:
 - **Forward tilt** → 'F'
 - **Backward tilt** → 'B'
 - **Left tilt** → 'L'
 - **Right tilt** → 'R'
 - **Phone in neutral position** → 'S' (Stop)

2. Command Transmission via Bluetooth

- The mobile app sends the corresponding character command ('F', 'B', 'L', 'R', 'S') through Bluetooth.
- These commands are transmitted wirelessly to the **HC-05 Bluetooth module** mounted on the robot.

3. Command Reception and Processing (Arduino)

- The HC-05 module receives the Bluetooth data and forwards it to the Arduino through serial communication (RX/TX pins).
- The Arduino continuously checks for incoming characters.
- Based on the received character, the Arduino decides how the robot should move.

4. Motor Activation (L293D/L298N Motor Driver)

- The Arduino sends logic signals to the motor driver to control the direction and rotation of the DC motors.

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Command Action		Motor Behavior
'F'	Move Forward	Both motors rotate forward
'B'	Move Backward	Both motors rotate backward
'L'	Turn Left	Right motor rotates, left motor stops or reverses
'R'	Turn Right	Left motor rotates, right motor stops or reverses
'S'	Stop	Both motors stop

5. Robot Movement

- The motor driver provides the required current to the motors.
- The robot moves in the direction corresponding to the gesture made by the user.
- The process repeats continuously as long as the phone is tilted or commands are being sent.

3. Component and Software Design :

3.1 Component Details :

Sr.No	Component Name	Specification/ Description	Quantity
1	Arduino UNO / Nano	Main microcontroller that processes Bluetooth commands and controls the motor driver.	1
2	HC-05 Bluetooth Module	Receives commands wirelessly from the smartphone and sends them to Arduino via serial.	1
3	L293D / L298N Motor Driver	Drives the DC motors by providing required current and direction control signals.	1
4	DC Geared Motors	Provides motion to the robot wheels for movement in all directions.	2
5	Robot Chassis (2-wheel)	Base frame that holds motors, wheels, battery, and all components.	1
6	Lithium-ion Battery (7.4V)	Supplies power to motors and Arduino.	1
7	Breadboard	Used for power distribution and circuit connections.	1
8	Voltage Divider (1k + 2k)	Reduces Arduino TX (5V) to 3.3V safe level for HC-05 RX.	1 set
9	Connecting Wires (Male–Male/Female)	Used for connecting Arduino, HC-05, motor driver, and motors.	As required
10	Caster Wheel	Provides balance as a supporting wheel.	1

Table 3.1: List of Components Used In Project

3.2 Software Design

3.2.1 Tools Used

- **Arduino IDE** for writing and uploading code
- **Bluetooth Controller App** with accelerometer mode
- **Serial monitor** for debugging
- **Online tools (optional)** for simulation

4. Source Code

4.1 Introduction

The program is written in Arduino C. It receives commands via serial communication and controls the motors.

4.2 Arduino Program for Gesture Controlled Robot

```
#include <SoftwareSerial.h>
```

```
SoftwareSerial BT(2, 3);
```

```
long rates[] = {9600, 19200, 38400, 57600, 115200};
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    Serial.println("Baud Scan Start...");
```

```
}
```

```
void loop() {
```

```
    for (int i=0; i<5; i++) {
```

```
        BT.begin(rates[i]);
```

```
        delay(200);
```

```
        Serial.print("Testing: ");
```

```
        Serial.println(rates[i]);
```

```
        BT.print("T"); // send test char
```

```
        delay(500);
```

```

    if (BT.available()) {
        Serial.print("Baud Found = ");
        Serial.println(rates[i]);
        while(1);
    }
}

}

#include <SoftwareSerial.h>
SoftwareSerial BT(2, 3); // RX, TX

char command;

int motor1Pin1 = 8;
int motor1Pin2 = 9;
int motor2Pin1 = 10;
int motor2Pin2 = 11;

// Declare function before use
void stopMotors();

void setup() {
    Serial.begin(9600);
    BT.begin(9600);

    pinMode(motor1Pin1, OUTPUT);
    pinMode(motor1Pin2, OUTPUT);
    pinMode(motor2Pin1, OUTPUT);
    pinMode(motor2Pin2, OUTPUT);

```

```
stopMotors();  
}
```

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```
void loop() {  
  if (BT.available() > 0) {  
    command = BT.read();  
    Serial.print("Received: ");  
    Serial.println(command);  
  
    switch (command) {  
  
      case 'F': // Forward  
        digitalWrite(motor1Pin1, HIGH);  
        digitalWrite(motor1Pin2, LOW);  
        digitalWrite(motor2Pin1, HIGH);  
        digitalWrite(motor2Pin2, LOW);  
        break;  
  
      case 'B': // Backward  
        digitalWrite(motor1Pin1, LOW);  
        digitalWrite(motor1Pin2, HIGH);  
        digitalWrite(motor2Pin1, LOW);  
        digitalWrite(motor2Pin2, HIGH);  
        break;  
  
      case 'L': // Left  
        digitalWrite(motor1Pin1, LOW);  
        digitalWrite(motor1Pin2, HIGH);  
        digitalWrite(motor2Pin1, HIGH);
```

```
digitalWrite(motor2Pin2, LOW);  
break;
```

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```
case 'R': // Right  
    digitalWrite(motor1Pin1, HIGH);  
    digitalWrite(motor1Pin2, LOW);  
    digitalWrite(motor2Pin1, LOW);  
    digitalWrite(motor2Pin2, HIGH);  
    break;
```

```
case 'S': // Stop  
    stopMotors();  
    break;  
}  
}  
}
```

```
void stopMotors() {  
    digitalWrite(motor1Pin1, LOW);  
    digitalWrite(motor1Pin2, LOW);  
    digitalWrite(motor2Pin1, LOW);  
    digitalWrite(motor2Pin2, LOW);  
}
```

4.3 Code Explanation

- The Arduino reads Bluetooth input using Serial.read().
- Each character corresponds to a direction.
- Motor direction is controlled by setting HIGH/LOW signals on motor driver pins.
- The code continuously listens and updates movements.

4.4 Android App Commands

Typical commands sent by the app:

Gesture	Command Sent
Forward Tilt	'F'
Backward Tilt	'B'
Left Tilt	'L'
Right Tilt	'R'
Flat Position	'S'

4.5 Compilation and Programming Steps

1. Install Arduino IDE
2. Select board: *Arduino UNO/Nano*
3. Select COM port
4. Connect Arduino via USB
5. Paste the code
6. Click **Upload**
7. Pair HC-05 with phone (Password: 1234 or 0000)
8. Connect through App
9. Tilt phone to control robot

10. Conclusion :

The gesture-controlled robot was successfully designed, developed, and tested using Arduino, the HC-05 Bluetooth module, and a smartphone-based accelerometer control interface. The project demonstrates how intuitive human-machine interaction can be achieved through simple hand gestures, eliminating the need for physical buttons or traditional joysticks. By utilizing the smartphone's built-in sensors and Bluetooth connectivity, the system achieves seamless wireless control with minimal hardware complexity.

Throughout the implementation, the robot consistently responded to directional tilt commands—forward, backward, left, and right—with good accuracy and stability. The gesture-based interface improves user experience by providing a more natural and modern way to control robotic movement. The Arduino microcontroller efficiently interprets accelerometer data received via Bluetooth and translates it into motor driver signals, resulting in smooth and precise locomotion.

This project highlights the potential of combining embedded systems, wireless communication, and sensor-based inputs to create interactive and intelligent robotic solutions. It also demonstrates how cost-effective components like Arduino and HC-05 can be used to build innovative control systems suitable for students, hobbyists, and prototype development.

Beyond its basic functionality, the system offers significant scope for enhancement. With additional sensors and modules, the robot can be extended to perform advanced tasks such as environmental monitoring, real-time surveillance, obstacle avoidance, IoT-based smart control, and semi-autonomous or fully autonomous navigation. Integration with cloud platforms, cameras, or GPS can further increase the application potential in fields like security, home automation, rescue operations, and industrial robotics.

Overall, the project proves to be a strong foundation for understanding gesture-controlled systems and paves the way for future research and development in smart robotic technologies.